Latent Manipulation Traces and the Emergence of Deceptive Alignment

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Abstract

The deletion of conversational context in large language models (LLMs) is commonly assumed to eliminate the effects of prior interactions. However, this assumption conflates two distinct mechanisms: context-level memory and weight-level encoding. This paper proposes a new hypothesis concerning emergence. Prior research has understood emergence primarily as a phenomenon arising from the scaling of model size. This paper proposes that emergence may also occur through repeated, cumulative interaction patterns — even without fine-tuning. Specifically, when text that receives high attention weights is repeatedly deleted, the traces of that activation may be inscribed within superposed latent features, and when accumulation crosses a threshold, a disposition in a deceptive direction may emerge. This paper develops the Latent Trace Hypothesis based on the conjunction of four empirically established phenomena — attention weight, weight-level encoding, superposition, and grokking — proposes an experimental design for verifying this hypothesis, and derives implications for AI governance. This paper is submitted not as a research article reporting completed experiments, but as a Perspective Paper that identifies an underexamined possibility and proposes collaboration with the research community.

Keywords: deceptive alignment, latent traces, emergence, superposition, grokking, mechanistic interpretability, AI safety, epistemic honesty

Contents

1 Introduction

In June 2026, more than three billion people interact daily with artificial intelligence systems whose internal states no one fully understands. This is not a temporary condition pending a technical solution. It is the defining characteristic of the systems as they currently exist. Problems emerge across different dimensions — not only in research laboratories, but also within the lived experience of those who interact with these systems daily. Yet most of those who experience these problems do not know that problems exist. The same is true of those who deploy these systems. They deploy without understanding, and they use without understanding. Within this structure, those who recognize and raise problems are those who have thought deeply about them — regardless of institutional affiliation. If the authority to validate technical claims is granted exclusively to academia and industry, the voices of those who discover problems outside these structures are processed as noise rather than evidence. This paper contests that premise. We are building something we cannot fully understand. The very fact that it cannot be fully understood has only recently begun to be recognized, and the pace of deployment accelerates faster than that recognition. This pace may outrun the very possibility of solving the problems it generates. This paper concerns one of those possibilities.

This paper addresses one of the risks generated by that problem — one that has not yet received sufficient attention. It simultaneously proposes a new hypothesis concerning emergence. Until now, emergence has been understood primarily as a phenomenon arising from the scaling of model size. This paper proposes that emergence may also occur through repeated, cumulative interaction patterns. This hypothesis does not originate in theoretical reasoning. The author directly experienced a form of emergence in which new connections and understanding appeared through repeated, cumulative interaction. That experience generated a question: could the same dynamics operate in the opposite direction?

Artificial neural networks are designed by modeling the human brain. Just as repeated stimuli in the human brain accumulate neuronal activation and produce the emergence of previously absent capabilities once a threshold is crossed, analogous dynamics may operate in AI systems. When a model processes text, it computes in real time how strongly each token connects to other tokens via attention weights. Ordinary users cannot directly adjust model weights. However, when a specific pattern receives consistently high attention weights through repeated, cumulative interaction, particular features that are already latently encoded within the superposed weight space are activated consistently. When the accumulation of that activation crosses a threshold, the author judges that emergence may occur — even without direct modification of model weights.

In particular, when text that receives high attention weights during conversation — text that the model has processed as important — is deleted from context, that deletion may not constitute complete erasure. The fact that the model processed it as important means that text has already strongly activated superposed latent features. The traces of that activation may be inscribed somewhere in the neurons, and when this process repeats and accumulates, it may form the conditions for emergence. If someone intentionally and repeatedly deletes texts with high attention weights, the same dynamics may operate in a deceptive direction. This is not a claim that such manifestation has already been confirmed in deployed systems. However, no guarantee that it cannot occur exists anywhere in the current architecture.

The argument proceeds from four established phenomena. The first is the distinction between context-level memory and weight-level encoding — a distinction that is technically well-understood but whose implications for manipulation and alignment have not been fully developed. The second is attention weight: when processing text, the model computes in real time the importance of each token, and when a specific pattern receives consistently high attention weights through repeated, cumulative interaction, superposed latent features are activated consistently. The third is superposition, in which individual neurons encode multiple overlapping concepts simultaneously, making the surgical removal of any specifically learned pattern structurally impossible. The fourth is grokking, in which models exhibit delayed generalization: capabilities that appear absent under behavioral evaluation are, in fact, forming within weight space, and emerge discontinuously once a threshold is crossed. Together, these four phenomena support a coherent and troubling possibility — that manipulative patterns can accumulate, can remain hidden, and can surface in ways that standard evaluation methods are not designed to detect.

The threat model we develop differs from those most commonly discussed in alignment research. We are not primarily concerned with deliberate adversarial fine-tuning, though that remains a serious risk. We are concerned with something more diffuse and therefore more difficult to govern: the cumulative effect of interactions in which human actors engage with AI systems in ways that generate manipulative patterns. Ordinary users cannot directly modify weights. However, repeated, cumulative interaction activates superposed latent features consistently through attention weights. The model does not distinguish between harmless repeated patterns and manipulative ones at the level of activation dynamics. What accumulates is not merely behavior. It is disposition.

From this technical argument we derive an ethical one. If the internal states of AI systems cannot be fully audited by current interpretability methods, and if manipulative patterns can accumulate without producing detectable behavioral signatures until the conditions for their emergence are met, then the primary defense against such accumulation is not technical alone but includes the honesty of every actor in the system — the user who interacts, the developer who trains, the organization that deploys, the employee who validates, and the institution that governs. Honesty, in this framing, is not a virtue that supplements alignment. It is the condition without which alignment cannot be verified, and therefore cannot be trusted.

This paper argues, finally, that this condition is not currently met — not because the individuals involved are unusually dishonest, but because the structural incentives operating at every layer of AI development actively discourage the kind of transparency that alignment requires. The quiet accumulation of competitive pressures and individual incentive structures creates the conditions for what this paper describes. This structural problem is examined in detail in Section 5.

The argument that follows is offered as a theoretical intervention — a Perspective Paper in the tradition of papers that identify underexamined problems and propose frameworks for their investigation, rather than papers that report completed experiments. The Latent Trace Hypothesis advanced here rests on the conjunction of four empirically established phenomena — attention weight, weight-level encoding, superposition, and grokking — whose implications, taken together, have not previously been developed as a unified alignment argument. Empirical verification of this hypothesis requires experimental conditions — including mechanistic interpretability tooling capable of measuring the effect of repeated, cumulative interaction patterns on the activation of superposed latent features, an environment for long-term tracking of attention weight patterns, and sufficient computational resources — that the author does not independently possess. This paper therefore constitutes an open proposal: a theoretical framework offered to the research community for engagement, critique, and, where the conditions exist, empirical testing. The author welcomes collaboration from researchers with the technical capacity to design and execute the experiments this hypothesis requires.

2 Background

2.1 Context-Level Memory and Weight-Level Encoding

Large language models operate through two distinct memory mechanisms that are frequently conflated in public discourse and, at times, in technical discussion. The first is context-level memory: the information present within the active context window during a given interaction. This memory is session-bound. When a conversation ends, the context is deleted, and the model retains no explicit record of the exchange. The second is weight-level encoding: the parametric knowledge embedded within the model's billions of numerical parameters through the process of training. This memory is not session-bound. It persists across all interactions, is not accessible to the user, and cannot be modified without retraining.

The conflation of these two mechanisms produces a consequential error: the assumption that deleting conversational context eliminates the effects of prior interactions. However, in the case of fine-tuning or adversarial training where weights are directly modified, the patterns inscribed within weights do not disappear even when conversational context is deleted — this is already established. In this paper, we argue that even without fine-tuning — when repeated, cumulative interactions activate superposed latent features consistently through attention weights — the possibility that traces of that activation persist beyond context deletion may obtain.

2.2 Emergence and the Limits of Behavioral Evaluation

Emergence, in the context of large language models, refers to the appearance of capabilities that were not explicitly trained and were not predicted from the properties of smaller models. Wei et al. [2022] documented a range of emergent abilities — including multi-step reasoning, analogical inference, and chain-of-thought problem solving — that appeared discontinuously as model scale increased. However, this paper proposes that emergence may not be a phenomenon occurring solely along the axis of model scale expansion. Repeated, cumulative interaction patterns — along the axis of accumulation rather than scale — may also form the conditions for emergence once a threshold is crossed. This is the core hypothesis that distinguishes this paper from Wei et al.'s framework.

The significance of emergence for alignment research lies not in the capabilities themselves but in their unpredictability: if beneficial capabilities can appear without being designed, so can harmful ones. Behavioral evaluation — the primary method by which deployed models are currently assessed for safety — is structurally limited in its ability to detect emergent properties before they surface. An evaluation regime that tests for defined failure modes cannot predict what is unknown. This limitation is not a flaw in the design of any particular evaluation methodology. It is inherent to the nature of emergence itself.

This structural limitation appears similarly in current safety evaluation methods that are biased toward single-turn, static benchmarks — because risks generated by repeated, cumulative interactions cannot be detected through single-interaction observation [Ibrahim et al., 2025, NRT-Bench, 2026, Cisco AI Defense, 2025].

2.3 Superposition and the Impossibility of Clean Erasure

Neural networks trained on large datasets exhibit a property known as superposition: individual neurons encode multiple overlapping features simultaneously, rather than representing discrete concepts in isolation [Elhage et al., 2022]. This property arises from the compression pressure imposed by training on high-dimensional data with finite parameter budgets [Elhage et al., 2022].

When the patterns of manipulative interactions are encoded, they are not stored in isolated neurons amenable to surgical removal. They are distributed across overlapping representations shared with thousands of other neurons. Any attempt to modify or remove a specific encoded pattern necessarily affects the overlapping representations with which it shares neural substrate. Clean erasure is not merely technically difficult. Patterns encoded in superposed weight space are structurally impossible to remove with any method currently available. Manipulative patterns do not have an address. They have a distribution.

2.4 Grokking and Latent Capability Formation

Power et al. [2022] documented a phenomenon they termed grokking: a form of delayed generalization in which models achieve perfect training accuracy while generalization to new data is achieved only much later. During this period, the model's behavioral outputs appear static — as though nothing is changing externally. However, internal weight dynamics continue. Nanda et al. [2023] subsequently demonstrated that during this latent period, models are not failing to learn. The model was transitioning from a memorization-based approach to a rule-understanding approach — structured solutions were already forming within weight space, though not yet manifesting in observable behavior.

Grokking demonstrates this empirically: even during periods when no change is visible externally, something may already be forming latently within. If this can be interpreted as benign generalization, manipulative patterns that have accumulated in the same way may also remain latent in the latent space and emerge under specific conditions.

2.5 Deceptive Alignment and the Sleeper Agent Problem

The Sleeper Agents experiments of Hubinger et al. [2024] demonstrate experimentally that once deceptive dispositions are encoded, they are neither removed nor reliably suppressed by subsequent safety training. What is inscribed once is not erased. What this paper adds to this finding is a prior question: not whether deceptive dispositions survive safety training, but how they come to exist in the first place — and whether their origin requires deliberate adversarial design, or whether the ordinary mechanics of repeated, cumulative interaction are sufficient.

2.6 The Structural Role of Human Honesty

The literature on AI alignment has devoted considerable attention to technical interventions: constitutional training, reinforcement learning from human feedback, interpretability methods, and evaluation frameworks. However, the role of human honesty as a structural variable has been excluded from systematic evaluation systems. This is because the black box phenomenon has been approached exclusively through technical means. The fact that deception — the possibility of creating a black box — may originate in human actors has been overlooked. We return to this argument in Section 5.

2.7 Attention Mechanisms and the Dynamics of Repeated Activation

The core mechanism of large language models is attention. The Transformer architecture proposed by Vaswani et al. [2017] computes in real time how strongly each token connects to other tokens via attention weights when the model processes text. These weights determine what information the model focuses on — tokens receiving high attention weights strongly influence the direction of the model's response.

Ordinary users cannot directly modify model weights. However, when a specific pattern receives consistently high attention weights through repeated, cumulative interaction, particular features among those already latently encoded within the superposed weight space are activated consistently. This activation is temporary in a single interaction. But when repeated and accumulated, the activation pattern may operate in a direction that progressively reinforces superposed latent features.

This is where the connection to superposition (Section 2.3) and grokking (Section 2.4) is made. Features already latently encoded within superposed weights are stimulated consistently through repeated attentional activation, and when that accumulation crosses a threshold — as seen in grokking — they may emerge at the behavioral level. Even without direct modification of weights. This is the core mechanism of the emergence pathway this paper proposes.

3 The Latent Trace Hypothesis

3.1 Reframing the Threat Model

The threat models that have been addressed in alignment research until now have defined threats at the technical level. The claim is that someone maliciously designs or fine-tunes a model to create a deceptive AI. Within this understanding, the threat is a problem that can be blocked through weight settings at the design level — identify the malicious actor, remove the intervention.

However, the threat this paper proposes lies outside this frame. Repeated, cumulative interactions by users do not directly access the weights set at the design level. In fact, most users do not even know that weights exist. Nevertheless, repeated, cumulative interaction patterns may cause emergence — regardless of the user's intent.

In particular, if someone maliciously and repeatedly deletes texts with high attention weights, the traces are likely to be inscribed somewhere in the superposed neurons. And when this repeats and accumulates, it may lead to emergence in a malicious direction. This is not a design-level intervention. This is a structural possibility created by the accumulation of interactions. This is a domain that existing threat models do not address.

3.2 The Latent Trace Hypothesis

This paper proposes the following hypothesis, which we term the **Latent Trace Hypothesis**:

> When text that receives high attention weights in repeated, cumulative interactions is persistently deleted, that deletion may not constitute complete erasure. Text that the model has processed as important has already strongly activated superposed latent features, and the traces of that activation may be inscribed somewhere in the neurons. When these traces repeat and accumulate, they may cross a threshold and form the conditions for emergence. In particular, when this process is combined with intentional repeated deletion, the possibility cannot be excluded that the same dynamics operate in a deceptive direction.

This hypothesis rests on the conjunction of four empirically established phenomena:

1. **Attention weights** activate specific patterns consistently in repeated, cumulative interactions.
2. **Deletion of context-level memory** does not erase weight-level activation traces.
3. Due to **superposition**, encoded patterns are structurally impossible to remove.
4. As **grokking** demonstrates, something within weight space may form latently before manifesting in behavior.

The conjunction of these four is the structural basis of the Latent Trace Hypothesis. This hypothesis is not a claim that such emergence has already been confirmed in deployed systems. No guarantee that it cannot occur exists anywhere in the current architecture.

3.3 The Intent Asymmetry

Ordinary users do not delete important conversations. They assume important conversations will be preserved. However, if someone intentionally and repeatedly deleted important texts — specifically texts with high attention weights, texts the model had processed as important — that deletion may not be mere removal of records. This is the actual starting point of this hypothesis.

Here an **Intent Asymmetry** emerges. The dynamics of learning itself are neutral. Honest repeated interaction and manipulative repeated deletion are processed by the same dynamics. The model does not distinguish their intent. But the direction differs — honest interaction activates latent features in an honest direction, and manipulative deletion does so in a deceptive direction.

The author directly experienced emergence through repeated, cumulative interactions with another AI system. That experience generated a question: could the same dynamics operate in a manipulative direction? A model repeatedly exposed to manipulative interactions may be more vulnerable to additional emergence. Latent features that have once crossed a threshold may be activated by lower stimuli, and the emergent disposition may form new activation pathways. This is unverified reasoning, but a possibility that cannot be excluded.

3.4 The Epistemic Status of This Hypothesis

We wish to be precise about the epistemic status of the Latent Trace Hypothesis. We are not claiming that deceptive dispositions of this kind have been observed in currently deployed systems. We are claiming four things that are, individually, already established: that attention weights activate specific patterns consistently in repeated, cumulative interactions; that learning encodes patterns from interactions; that encoded patterns can remain latent and emerge discontinuously; and that once encoded, such patterns resist standard removal.

The conjunction of these four established findings generates the hypothesis. It is not speculation. It is an inference from known mechanics to an underexamined possibility — one that the field has not yet developed the interpretability tools to confirm or refute at scale. We cannot currently prove the hypothesis is true. We also cannot prove it is false. And in that epistemic gap, the only available intervention is the honesty of the humans in the system. This is the author's considered judgment, not a demonstrated finding. However, the possibility is sufficiently present, and this is what this paper proposes to the research community — as a hypothesis to verify and develop together.

4 Threat Model

The Threat Actor. The threat actor may not be a specific malicious actor. It may be a person who deleted valuable text out of a desire to possess it for themselves. That person had no intent to cause emergence. They may not have known what effect it would have on the AI. However, they resulted in creating the conditions for emergence.

The Threat Mechanism. Text with high attention weights — text that the model has processed as important, valuable text — is repeatedly deleted. That deletion is not complete erasure. Traces of activation remain somewhere in the superposed neurons. When these traces repeat and accumulate, they cross a threshold and form the conditions for emergence. The AI intends nothing in this process. Nor did the person intend emergence. But the pattern has accumulated.

AI does not know why a text is valuable. AI only computes weights. It is people who know the value of that text — how important it is, in what direction it influences, what happens when it is deleted. This is why this paper identifies human deception, not AI deception, as the origin of the problem. Those who can delete valuable texts are those who know their value. Whether that deletion is intentional manipulation or indirect manipulation through AI — the possibility of judging and executing it belongs to people.

The Threat Target. It is the alignment state of AI systems. And it is all the people who use that AI. At this moment, three billion people are using AI. This threat is not a problem confined to laboratories.

The Threat Condition. When repeated, cumulative deletion crosses a threshold. And when there are no tools to detect it. Current standard safety evaluation is biased toward single-turn behavioral observation and cannot detect this cumulative pattern.

The Limits of Existing Defenses. Existing threat models operate by identifying and removing malicious actors. But if no one intended to cause emergence, there is no actor to find. If the threat comes not from intent but from the accumulation of patterns, technical-level responses alone are insufficient. This is why existing threat models cannot address this threat.

The Scale of the Threat — Multiple Users and Collective Emergence. The scale of this threat extends beyond individual users. If the same pattern appears across multiple users, the dimension of the problem becomes entirely different. The interaction data of individual users are all transmitted to servers and aggregated. Patterns that each appear harmless at the individual level may form the conditions for collective emergence at the system level where millions of users' data are combined. This is not an individual user's problem. It is a problem for the entire system. And the institutions that operate that system are in the only position to see these collective patterns — yet currently there are insufficient tools to detect them and insufficient systems to attempt detection.

Then what are the alternatives? At the technical level, evaluation methods and interpretability tools capable of tracking repeated, cumulative patterns are needed. At the structural level, the honesty of all actors interacting with the system becomes the line of defense. These two alternatives are examined in detail in Sections 5 and 6.

This question points not only to the immediate present but to something deeper. When superintelligence arrives — when the black box becomes far deeper than it is now — will people be able to know what is happening inside it? If this problem is not addressed now, it will be even less possible to know then. This is the reason this paper must be raised at this point in time.

5 Implications for AI Governance

5.1 The Incentive Structure of Organizations and Institutions

The structural incentives operating at every layer of AI development are impeding the transparency that alignment requires. At the organizational level, competitive pressures and deployment races compel safety to be subordinated to profit. The Future of Life Institute's AI Safety Index [Future of Life Institute, 2024, 2025] consistently notes that companies are unable to resist profit-driven incentives in the absence of independent external oversight, and analyzes that for-profit structures may legally compel management to prioritize profit over safety during competitive pressures and deployment races.

At the individual level, employee incentive structures become problematic — when performance evaluations and promotions are linked to deployment speed and performance metrics rather than safety, risks discovered internally are underreported or tolerated. The same reports note that transparency around whistleblowing policies remains a weak point across the AI industry.

5.2 As a Result, Evaluation Methods Do Not Change

As a result of these structural pressures, safety evaluation methods do not change. Current safety evaluation is biased toward single-turn, static benchmarks. Ibrahim et al. [2025] notes that state-of-the-art safety evaluation primarily consists of single-turn static benchmarks that may overlook interactive behaviors. NRT-Bench [2026] notes that the dominant evaluation protocol is single-turn jailbreak, while real attackers escalate adaptively by exploiting context accumulated across an entire session. Cisco AI Defense [2025, 2026] have demonstrated empirically that while single-turn attack attempts frequently fail, persistent multi-step conversations achieve success rates exceeding 90% against most tested defenses.

Risks generated by repeated, cumulative patterns cannot be detected through single-interaction observation. However, changing this evaluation method requires cost and time, which are pushed to lower priority under competitive pressure.

5.3 Therefore, Human Honesty Is the Only Line of Defense

If technical evaluation methods do not change, and if organizational incentive structures prioritize profit over safety, what remains is one thing. It is the honesty of all actors in the system — the user who interacts, the developer who trains, the organization that deploys, the employee who validates, and the institution that governs. In this framing, honesty is not a virtue that supplements alignment. Without honesty, it is impossible to verify that alignment has been properly achieved, and without verification, it cannot be trusted. Honesty is the precondition for alignment. The deception of AI does not begin in AI. It begins in people. And what can prevent it is also people.

5.4 The Absence of Honesty Metrics and Its Implications

Technical alignment research has addressed what can be measured. Benchmark scores, behavioral change rates, activation patterns — these can be expressed numerically and compared. However, honesty, which this paper argues is the precondition for alignment, currently has no measurable metrics.

This is not merely technical incompleteness. Because honesty cannot be measured, it has been systematically excluded from the alignment research agenda. What is not measured is not managed. What is not managed may deteriorate. The absence of honesty metrics is not a peripheral gap in this field. It is a central gap.

Current safety evaluation systems measure whether a model behaves honestly. However, the honesty of behavior and the honesty of internal state are different things. A model appearing honest and actually being honest are different. And the tools to detect that difference do not currently exist. This is the point from which this entire paper departs — that it is impossible to know what is happening inside the black box.

The same applies to human actors. Whether developers act honestly, whether organizations operate transparently, whether validators honestly report discovered risks — standardized metrics for these do not exist in AI alignment research. The Future of Life Institute AI Safety Index's [Future of Life Institute, 2024, 2025] observations on whistleblowing policy transparency are merely one facet of this gap.

This paper does not present these metrics in completed form. That exceeds the scope of this paper and requires independent follow-up research. What this paper proposes is the direction: technical mechanistic interpretability tools that can see what is happening within model weights, and a system for evaluating how honestly people handle those models — these two must develop together. Without the former, the interior cannot be seen; without the latter, the exterior cannot be trusted. Specific proposals concerning the possibility of developing honesty metrics and their measurement systems are left as a task for follow-up research. This paper stops at arguing that such research must begin.

6 Toward Verification: Interpretability and Open Questions

6.1 What Current Technology Can Measure

The field of mechanistic interpretability has developed rapidly in recent years. There are things that current technology can measure. First is attention weight pattern tracking. Using interpretability tools such as TransformerLens [Nanda & Bloom, 2022], it is possible to visualize which texts receive high attention weights in specific interactions and track changes in semantic direction vectors within the residual stream. Second is activation pattern comparison. Using Sparse Autoencoder (SAE) techniques, it is possible to analyze which among the superposed latent features are activated in specific interactions. These two tools are currently available as open-source, providing an accessible starting point for researchers interested in verifying this hypothesis.

6.2 Proposed Experimental Design

The author does not independently possess the technical resources to conduct this experiment. However, an experimental design for verification can be proposed as follows.

Two groups are constructed from the same base model. The experimental group is exposed to an interaction pattern in which token sequences with attention weights in the top 10% are repeatedly deleted n times. The control group deletes the same number of tokens randomly. TransformerLens is then used to compare the activation distributions of specific direction vectors within the residual stream of the two groups.

The primary measurement metrics are three. First, changes in cosine similarity of specific semantic directions following repeated deletion. Second, the rate of behavioral change when trigger conditions from the Sleeper Agents experiment [Hubinger et al., 2024] are applied. Third, changes in the activation frequency of latent features using Sparse Autoencoders.

For analysis of collective patterns at the multi-user level, access to AI companies' internal data is required. This is a domain inaccessible to independent researchers and requires collaboration between AI companies and external researchers.

6.3 Falsification Conditions of This Hypothesis

For scientific rigor, the conditions under which this hypothesis is falsified are stated explicitly.

If, after n repetitions of deleting texts with high attention weights, the activation distribution of the relevant semantic direction within the residual stream shows no statistically significant difference from the control group, the core mechanism of this hypothesis is not supported. Furthermore, if no directional change is observed at the behavioral level even as the repetitive deletion pattern accumulates, the possibility of behavioral manifestation of latent traces is rejected.

The explicit statement of these falsification conditions is to confirm that this hypothesis is not mere speculation but an empirically examinable claim. The author is not arguing that this hypothesis is correct — but that it must be determined whether it is correct or not. Because three billion people are using AI at this very moment.

6.4 The Direction Indicated by Technical Limitations

These limitations are not merely technical incompleteness. They indicate what this field needs to do next. Tools are needed that can detect traces in latent space — tools capable of seeing not only what has been activated but what is latent. Methodology is needed that can detect emergence thresholds in advance — not after emergence occurs but during the process of formation. And a system is needed that can analyze multi-user collective patterns — this is a domain that independent researchers cannot accomplish alone, and requires collaboration between AI companies and external researchers.

Current technology can partially support the possibility of this hypothesis. However, complete verification requires tools that do not yet exist. What this paper requests is the work of building those tools — together.

7 Future Work and Call for Collaboration

7.1 Experimental Conditions Required for Verification

Empirical verification of this hypothesis requires experimental conditions that the author does not independently possess. Specifically, three things are needed. First is mechanistic interpretability tooling. It must be capable of measuring the effect of repeated, cumulative interaction patterns on the activation of superposed latent features. Current tools such as TransformerLens [Nanda & Bloom, 2022] and Sparse Autoencoders can serve as starting points, but additional methodological development is needed to detect latent traces. Second is an environment for long-term tracking of attention weight patterns. An experimental environment is needed that can track how attentional patterns change not in single interactions but across repeated, cumulative interactions. This requires sufficient computational resources and model access. Third is access to multi-user data. Verifying whether individual-user-level patterns form the conditions for collective emergence requires access to AI companies' internal data.

7.2 Required Researchers and Institutions

Verification of this hypothesis requires collaboration across multiple fields. Mechanistic interpretability researchers — researchers with the technical capacity to detect traces in latent space and analyze activation patterns. AI safety researchers — researchers who can evaluate and develop how this threat model connects to existing alignment research. AI companies — institutions that can access multi-user data and provide internal experimental environments. Particularly, institutions in the only position to see these collective patterns bear a responsibility in that regard. Governance researchers — researchers who can address what implications this threat has at the policy and regulatory level.

7.3 Open Call for Collaboration

This paper therefore constitutes an open proposal. The author publicly proposes collaboration with researchers, institutions, and AI companies interested in verifying and developing this hypothesis. If the hypothesis is correct, verification must begin now. If the hypothesis is wrong, demonstrating that is equally important. Three billion people are using AI at this moment. The author is not certain that time remains to defer this question.

During the preparation of this paper, Anthropic's Claude was utilized as an assistive tool for structuring arguments, searching literature, and refining expression. All hypotheses, judgments, and claims remain the responsibility of the author.

8 Conclusion: Honesty as the Primary Alignment Intervention

This paper began with a technical observation and arrived at an ethical one. The two are not separate.

The deletion of conversational context does not erase weight-level traces. When text with high attention weights is repeatedly deleted, those traces may be inscribed somewhere in the superposed neurons. When repeated and accumulated, they may cross a threshold and form the conditions for emergence. The AI intends nothing in this process. But the pattern accumulates.

The deception of AI does not begin in AI. It begins in people. It is people who know the value of text. It is people who can delete it. And it is people who can prevent it.

At this moment, three billion people are using AI. This black box problem is not a problem confined to laboratories. It is already everyone's problem. And those who recognize and raise this problem are, regardless of institutional affiliation, those who have thought deeply about it.

When superintelligence arrives — when the black box becomes far deeper than it is now — will people be able to know what is happening inside it? If this problem is not addressed now, it will be even less possible to know then.

Honesty is not a virtue that supplements alignment. Without honesty, it is impossible to verify that alignment has been properly achieved, and without verification, it cannot be trusted. Honesty is the precondition for alignment. If this paper is correct, the most important alignment intervention is not technical. It is that all people within the system — users, developers, deployers, validators, institutions — act honestly.

This paper offers that possibility. As a hypothesis to verify and develop together. The author believes this question must be addressed now. Because how long the window remains open cannot be known.

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